

LOSO (glove_ml_explorer) Experiment SA1

Static · full sensors (flex+ypr+accel, both hands), INCLUDING Double_Nothing

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Experiment configuration

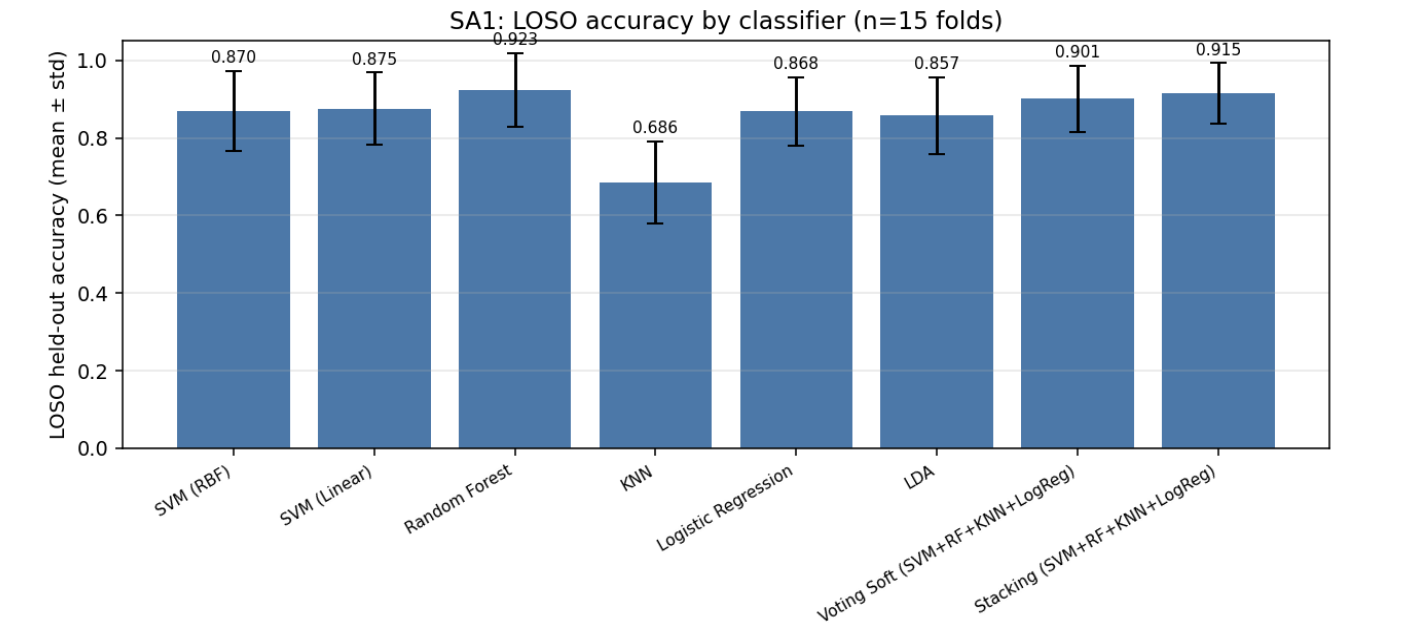
Class set	7 classes including Double_Nothing
Classes	Double_Closed_Fist, Double_Nothing, Double_Okay, Double_Open_Palm, Double_Phone, Double_Pistol, Double_Spiderman
Sensor modalities	flex, YPR, accel
Hands	left, right
Segments	thumb, index, middle, ring, pinky, wrist
Sensor channels	152
Sequence length	90 steps (resampled)
Filter	off
Feature mode	stats+fft
Normalisation	minmax
Augmentation	off
Inner CV folds	5
LOSO subjects	15: 10_Josh, 11_Mansh, 12_Marcus, 13_Nat, 14_StephenV2, 16_Bella, 17_Raquel, 1_Alan, 2_Alex, 3_Anghad, 4_Daniel, 5_Georgia, 6_Harry, 7_Henry, 8_Jestin

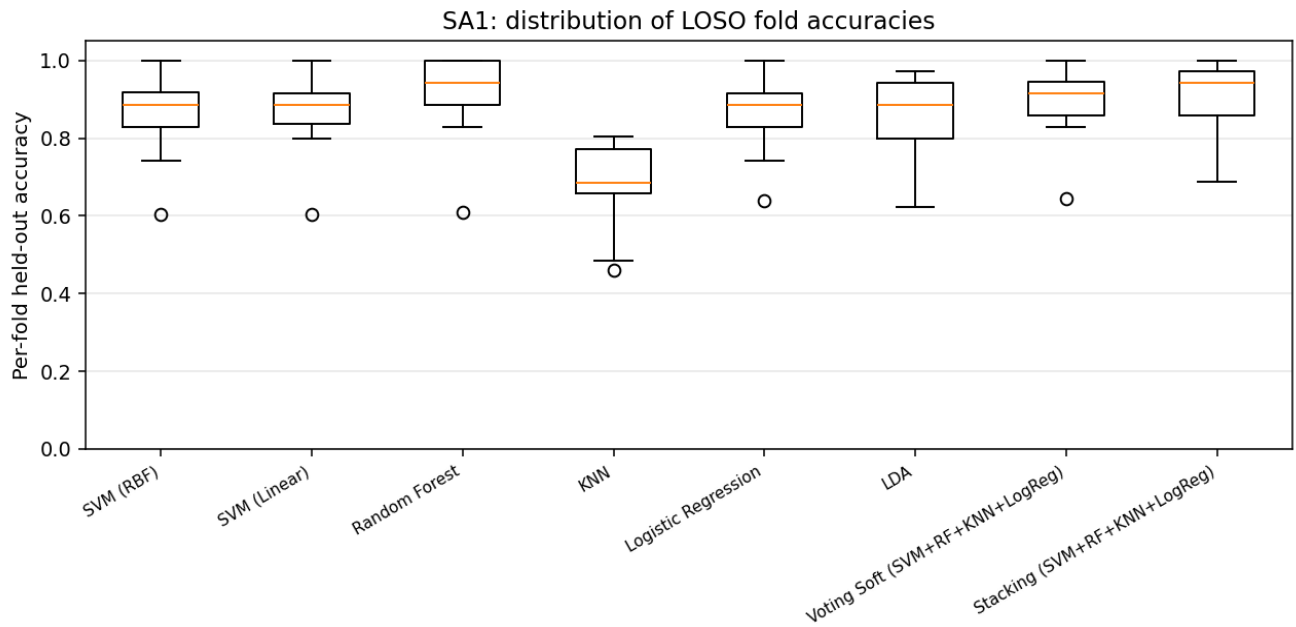
LOSO summary across all subjects

Classifier	Params*	LOSO mean	LOSO std	Min fold	Max fold	Inner CV mean
SVM (RBF)	—	0.8698	0.1032	0.6028	1.0000	0.9393
SVM (Linear)	172,389	0.8749	0.0934	0.6028	1.0000	0.9418
Random Forest	23,492	0.9233	0.0955	0.6099	1.0000	0.9456
KNN	—	0.6857	0.1061	0.4610	0.8056	0.7636
Logistic Regression	57,463	0.8685	0.0885	0.6383	1.0000	0.9272
LDA	57,463	0.8573	0.0989	0.6241	0.9714	0.8943
Voting Soft (SVM+RF+KNN+LogReg)	—	0.9007	0.0862	0.6454	1.0000	0.9484
Stacking (SVM+RF+KNN+LogReg)	—	0.9146	0.0781	0.6879	1.0000	0.9545

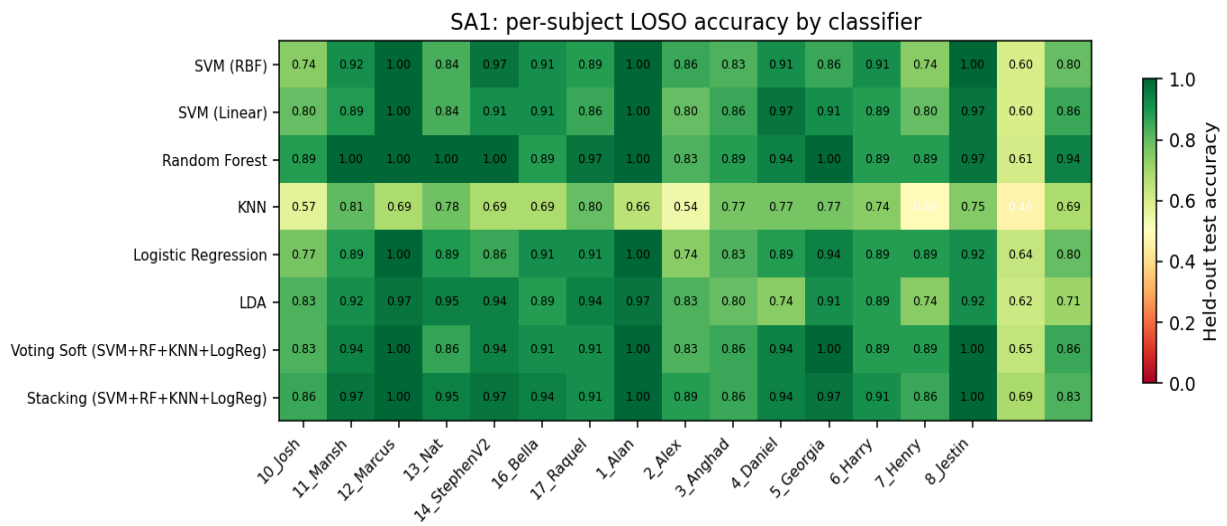
*Params: linear models use coef_+intercept size; tree ensembles use total node count across trees. SVM-RBF / KNN / Voting / Stacking are non-parametric or composite, so they show as —.

Best classifier by LOSO mean: **Random Forest**





Per-subject LOSO accuracy heatmap



Per-fold detail (held-out subject = each row)

Held-out subj	Train trials	SVM (RBF)	SVM (Linear)	Random Forest	KNN	Logistic Regression	LDA	Voting Soft (SVM+RF+KNN+LogReg)	Stacking (SVM+RF+KNN+LogReg)
10_Josh	670	0.743	0.800	0.886	0.571	0.771	0.829	0.829	0.857
11_Mansh	669	0.917	0.889	1.000	0.806	0.889	0.917	0.944	0.972
12_Marcus	670	1.000	1.000	1.000	0.686	1.000	0.971	1.000	1.000
13_Nat	668	0.838	0.838	1.000	0.784	0.892	0.946	0.865	0.946
14_StephenV2	670	0.971	0.914	1.000	0.686	0.857	0.943	0.943	0.971
16_Bella	670	0.914	0.914	0.886	0.686	0.914	0.886	0.914	0.943
17_Raquel	670	0.886	0.857	0.971	0.800	0.914	0.943	0.914	0.914
1_Alan	670	1.000	1.000	1.000	0.657	1.000	0.971	1.000	1.000
2_Alex	670	0.857	0.800	0.829	0.543	0.743	0.829	0.829	0.886
3_Anghad	670	0.829	0.857	0.886	0.771	0.829	0.800	0.857	0.857
4_Daniel	670	0.914	0.971	0.943	0.771	0.886	0.743	0.943	0.943
5_Georgia	670	0.857	0.914	1.000	0.771	0.943	0.914	1.000	0.971
6_Harry	670	0.914	0.886	0.886	0.743	0.886	0.886	0.886	0.914
7_Henry	670	0.743	0.800	0.886	0.486	0.886	0.743	0.886	0.857
8_Jestin	669	1.000	0.972	0.972	0.750	0.917	0.917	1.000	1.000
Stephen	564	0.603	0.603	0.610	0.461	0.638	0.624	0.645	0.688
Val	670	0.800	0.857	0.943	0.686	0.800	0.714	0.857	0.829

Classifier: SVM (RBF)

Params (best-effort): —

LOSO mean accuracy: 0.8698 ± 0.1032 (min 0.6028, max 1.0000)

Per-fold per-class held-out accuracy

Held-out	Double_Closed_Fist	Double_Nothing	Double_Okay	Double_Open_Palm	Double_Phone	Double_Pistol	Double_Spiderman	Overall
10_Josh	1/5	3/5	3/5	4/5	5/5	5/5	5/5	0.743
11_Mansh	3/6	5/5	5/5	5/5	5/5	5/5	5/5	0.917
12_Marcus	5/5	5/5	5/5	5/5	5/5	5/5	5/5	1.000
13_Nat	6/6	5/5	5/5	5/5	0/5	5/6	5/5	0.838
14_StephenV2	5/5	4/5	5/5	5/5	5/5	5/5	5/5	0.971
16_Bella	5/5	2/5	5/5	5/5	5/5	5/5	5/5	0.914
17_Raquel	5/5	4/5	2/5	5/5	5/5	5/5	5/5	0.886
1_Alan	5/5	5/5	5/5	5/5	5/5	5/5	5/5	1.000
2_Alex	5/5	5/5	5/5	5/5	0/5	5/5	5/5	0.857
3_Anghad	4/5	1/5	5/5	4/5	5/5	5/5	5/5	0.829
4_Daniel	4/5	4/5	5/5	5/5	4/5	5/5	5/5	0.914
5_Georgia	3/5	5/5	3/5	4/5	5/5	5/5	5/5	0.857
6_Harry	5/5	3/5	4/5	5/5	5/5	5/5	5/5	0.914
7_Henry	5/5	5/5	2/5	4/5	3/5	2/5	5/5	0.743
8_Jestin	6/6	5/5	5/5	5/5	5/5	5/5	5/5	1.000
Stephen	2/20	14/21	12/20	12/20	11/20	16/20	18/20	0.603
Val	2/5	3/5	5/5	5/5	4/5	4/5	5/5	0.800
MEAN	0.788	0.792	0.847	0.929	0.809	0.931	0.994	0.870

Classifier: SVM (Linear)

Params (best-effort): 172,389

LOSO mean accuracy: 0.8749 ± 0.0934 (min 0.6028, max 1.0000)

Per-fold per-class held-out accuracy

Held-out	Double_Closed_Fist	Double_Nothing	Double_Okay	Double_Open_Palm	Double_Phone	Double_Pistol	Double_Spiderman	Overall
10_Josh	1/5	5/5	3/5	4/5	5/5	5/5	5/5	0.800
11_Mansh	3/6	5/5	4/5	5/5	5/5	5/5	5/5	0.889
12_Marcus	5/5	5/5	5/5	5/5	5/5	5/5	5/5	1.000
13_Nat	6/6	5/5	5/5	5/5	0/5	5/6	5/5	0.838
14_StephenV2	5/5	4/5	5/5	5/5	5/5	3/5	5/5	0.914
16_Bella	5/5	2/5	5/5	5/5	5/5	5/5	5/5	0.914
17_Raquel	5/5	4/5	2/5	5/5	4/5	5/5	5/5	0.857
1_Alan	5/5	5/5	5/5	5/5	5/5	5/5	5/5	1.000
2_Alex	5/5	3/5	5/5	5/5	0/5	5/5	5/5	0.800
3_Anghad	4/5	1/5	5/5	5/5	5/5	5/5	5/5	0.857
4_Daniel	5/5	4/5	5/5	5/5	5/5	5/5	5/5	0.971
5_Georgia	5/5	4/5	4/5	4/5	5/5	5/5	5/5	0.914
6_Harry	5/5	3/5	4/5	5/5	4/5	5/5	5/5	0.886
7_Henry	5/5	3/5	2/5	5/5	4/5	4/5	5/5	0.800
8_Jestin	6/6	4/5	5/5	5/5	5/5	5/5	5/5	0.972
Stephen	2/20	8/21	12/20	13/20	15/20	17/20	18/20	0.603
Val	3/5	2/5	5/5	5/5	5/5	5/5	5/5	0.857
MEAN	0.835	0.717	0.847	0.956	0.832	0.946	0.994	0.875

Classifier: Random Forest

Params (best-effort): 23,492

LOSO mean accuracy: 0.9233 ± 0.0955 (min 0.6099, max 1.0000)

Per-fold per-class held-out accuracy

Held-out	Double_Closed_Fist	Double_Nothing	Double_Okay	Double_Open_Palm	Double_Phone	Double_Pistol	Double_Spiderman	Overall
10_Josh	5/5	1/5	5/5	5/5	5/5	5/5	5/5	0.886
11_Mansh	6/6	5/5	5/5	5/5	5/5	5/5	5/5	1.000
12_Marcus	5/5	5/5	5/5	5/5	5/5	5/5	5/5	1.000
13_Nat	6/6	5/5	5/5	5/5	5/5	6/6	5/5	1.000
14_StephenV2	5/5	5/5	5/5	5/5	5/5	5/5	5/5	1.000
16_Bella	5/5	1/5	5/5	5/5	5/5	5/5	5/5	0.886
17_Raquel	5/5	4/5	5/5	5/5	5/5	5/5	5/5	0.971
1_Alan	5/5	5/5	5/5	5/5	5/5	5/5	5/5	1.000
2_Alex	5/5	4/5	5/5	5/5	0/5	5/5	5/5	0.829
3_Anghad	5/5	1/5	5/5	5/5	5/5	5/5	5/5	0.886
4_Daniel	5/5	3/5	5/5	5/5	5/5	5/5	5/5	0.943
5_Georgia	5/5	5/5	5/5	5/5	5/5	5/5	5/5	1.000
6_Harry	5/5	3/5	5/5	5/5	3/5	5/5	5/5	0.886
7_Henry	5/5	4/5	5/5	5/5	5/5	2/5	5/5	0.886
8_Jestin	6/6	4/5	5/5	5/5	5/5	5/5	5/5	0.972
Stephen	1/20	10/21	16/20	20/20	7/20	14/20	18/20	0.610
Val	4/5	5/5	5/5	5/5	4/5	5/5	5/5	0.943
MEAN	0.932	0.734	0.988	1.000	0.868	0.947	0.994	0.923

Classifier: KNN

Params (best-effort): —

LOSO mean accuracy: 0.6857 ± 0.1061 (min 0.4610, max 0.8056)

Per-fold per-class held-out accuracy

Held-out	Double_Closed_Fist	Double_Nothing	Double_Okay	Double_Open_Palm	Double_Phone	Double_Pistol	Double_Spiderman	Overall
10_Josh	4/5	1/5	0/5	3/5	2/5	5/5	5/5	0.571
11_Mansh	3/6	5/5	3/5	5/5	3/5	5/5	5/5	0.806
12_Marcus	4/5	3/5	2/5	4/5	1/5	5/5	5/5	0.686
13_Nat	6/6	4/5	5/5	3/5	0/5	6/6	5/5	0.784
14_StephenV2	3/5	1/5	5/5	2/5	3/5	5/5	5/5	0.686
16_Bella	3/5	0/5	4/5	5/5	2/5	5/5	5/5	0.686
17_Raquel	5/5	2/5	3/5	5/5	3/5	5/5	5/5	0.800
1_Alan	5/5	3/5	3/5	2/5	0/5	5/5	5/5	0.657
2_Alex	5/5	1/5	1/5	1/5	1/5	5/5	5/5	0.543
3_Anghad	4/5	1/5	5/5	4/5	3/5	5/5	5/5	0.771
4_Daniel	5/5	1/5	5/5	4/5	2/5	5/5	5/5	0.771
5_Georgia	3/5	5/5	3/5	5/5	1/5	5/5	5/5	0.771
6_Harry	5/5	1/5	5/5	3/5	2/5	5/5	5/5	0.743
7_Henry	4/5	1/5	3/5	0/5	1/5	3/5	5/5	0.486
8_Jestin	4/6	4/5	5/5	2/5	2/5	5/5	5/5	0.750
Stephen	2/20	3/21	10/20	10/20	5/20	17/20	18/20	0.461
Val	1/5	2/5	2/5	5/5	5/5	5/5	4/5	0.686
MEAN	0.733	0.420	0.665	0.653	0.379	0.968	0.982	0.686

Classifier: Logistic Regression

Params (best-effort): 57,463

LOSO mean accuracy: 0.8685 ± 0.0885 (min 0.6383, max 1.0000)

Per-fold per-class held-out accuracy

Held-out	Double_Closed_Fist	Double_Nothing	Double_Okay	Double_Open_Palm	Double_Phone	Double_Pistol	Double_Spiderman	Overall
10_Josh	1/5	4/5	4/5	3/5	5/5	5/5	5/5	0.771
11_Mansh	3/6	5/5	4/5	5/5	5/5	5/5	5/5	0.889
12_Marcus	5/5	5/5	5/5	5/5	5/5	5/5	5/5	1.000
13_Nat	6/6	5/5	5/5	5/5	2/5	5/6	5/5	0.892
14_StephenV2	4/5	4/5	5/5	4/5	5/5	3/5	5/5	0.857
16_Bella	5/5	2/5	5/5	5/5	5/5	5/5	5/5	0.914
17_Raquel	5/5	4/5	3/5	5/5	5/5	5/5	5/5	0.914
1_Alan	5/5	5/5	5/5	5/5	5/5	5/5	5/5	1.000
2_Alex	3/5	2/5	5/5	5/5	1/5	5/5	5/5	0.743
3_Anghad	4/5	1/5	5/5	4/5	5/5	5/5	5/5	0.829
4_Daniel	5/5	3/5	5/5	4/5	4/5	5/5	5/5	0.886
5_Georgia	5/5	5/5	4/5	4/5	5/5	5/5	5/5	0.943
6_Harry	5/5	4/5	5/5	5/5	2/5	5/5	5/5	0.886
7_Henry	5/5	2/5	4/5	5/5	5/5	5/5	5/5	0.886
8_Jestin	6/6	3/5	5/5	5/5	5/5	5/5	4/5	0.917
Stephen	2/20	7/21	15/20	15/20	17/20	16/20	18/20	0.638
Val	2/5	2/5	5/5	5/5	4/5	5/5	5/5	0.800
MEAN	0.788	0.678	0.915	0.915	0.850	0.955	0.982	0.868

Classifier: LDA

Params (best-effort): 57,463

LOSO mean accuracy: 0.8573 ± 0.0989 (min 0.6241, max 0.9714)

Per-fold per-class held-out accuracy

Held-out	Double_Closed_Fist	Double_Nothing	Double_Okay	Double_Open_Palm	Double_Phone	Double_Pistol	Double_Spiderman	Overall
10_Josh	1/5	4/5	5/5	4/5	5/5	5/5	5/5	0.829
11_Mansh	4/6	5/5	4/5	5/5	5/5	5/5	5/5	0.917
12_Marcus	5/5	5/5	5/5	5/5	4/5	5/5	5/5	0.971
13_Nat	6/6	4/5	5/5	5/5	5/5	5/6	5/5	0.946
14_StephenV2	5/5	4/5	5/5	4/5	5/5	5/5	5/5	0.943
16_Bella	5/5	2/5	5/5	5/5	4/5	5/5	5/5	0.886
17_Raquel	5/5	3/5	5/5	5/5	5/5	5/5	5/5	0.943
1_Alan	5/5	5/5	5/5	5/5	4/5	5/5	5/5	0.971
2_Alex	4/5	3/5	5/5	5/5	2/5	5/5	5/5	0.829
3_Anghad	3/5	1/5	5/5	5/5	4/5	5/5	5/5	0.800
4_Daniel	3/5	3/5	5/5	3/5	2/5	5/5	5/5	0.743
5_Georgia	5/5	4/5	4/5	5/5	4/5	5/5	5/5	0.914
6_Harry	5/5	3/5	4/5	5/5	4/5	5/5	5/5	0.886
7_Henry	4/5	1/5	4/5	3/5	5/5	4/5	5/5	0.743
8_Jestin	5/6	3/5	5/5	5/5	5/5	5/5	5/5	0.917
Stephen	2/20	13/21	9/20	20/20	13/20	13/20	18/20	0.624
Val	3/5	4/5	3/5	3/5	3/5	4/5	5/5	0.714
MEAN	0.776	0.672	0.897	0.906	0.815	0.946	0.994	0.857

Classifier: Voting Soft (SVM+RF+KNN+LogReg)

Params (best-effort): —

LOSO mean accuracy: 0.9007 ± 0.0862 (min 0.6454, max 1.0000)

Per-fold per-class held-out accuracy

Held-out	Double_Closed_Fist	Double_Nothing	Double_Okay	Double_Open_Palm	Double_Phone	Double_Pistol	Double_Spiderman	Overall
10_Josh	1/5	4/5	5/5	4/5	5/5	5/5	5/5	0.829
11_Mansh	4/6	5/5	5/5	5/5	5/5	5/5	5/5	0.944
12_Marcus	5/5	5/5	5/5	5/5	5/5	5/5	5/5	1.000
13_Nat	6/6	4/5	5/5	5/5	2/5	5/6	5/5	0.865
14_StephenV2	5/5	4/5	5/5	4/5	5/5	5/5	5/5	0.943
16_Bella	5/5	2/5	5/5	5/5	5/5	5/5	5/5	0.914
17_Raquel	5/5	4/5	4/5	5/5	4/5	5/5	5/5	0.914
1_Alan	5/5	5/5	5/5	5/5	5/5	5/5	5/5	1.000
2_Alex	5/5	4/5	5/5	5/5	0/5	5/5	5/5	0.829
3_Anghad	4/5	1/5	5/5	5/5	5/5	5/5	5/5	0.857
4_Daniel	5/5	3/5	5/5	5/5	5/5	5/5	5/5	0.943
5_Georgia	5/5	5/5	5/5	5/5	5/5	5/5	5/5	1.000
6_Harry	5/5	3/5	5/5	5/5	3/5	5/5	5/5	0.886
7_Henry	5/5	3/5	4/5	5/5	5/5	4/5	5/5	0.886
8_Jestin	6/6	5/5	5/5	5/5	5/5	5/5	5/5	1.000
Stephen	2/20	8/21	15/20	15/20	17/20	16/20	18/20	0.645
Val	2/5	3/5	5/5	5/5	5/5	5/5	5/5	0.857
MEAN	0.833	0.728	0.962	0.962	0.862	0.967	0.994	0.901

Classifier: Stacking (SVM+RF+KNN+LogReg)

Params (best-effort): —

LOSO mean accuracy: 0.9146 ± 0.0781 (min 0.6879, max 1.0000)

Per-fold per-class held-out accuracy

Held-out	Double_Closed_Fist	Double_Nothing	Double_Okay	Double_Open_Palm	Double_Phone	Double_Pistol	Double_Spiderman	Overall
10_Josh	1/5	5/5	5/5	4/5	5/5	5/5	5/5	0.857
11_Mansh	5/6	5/5	5/5	5/5	5/5	5/5	5/5	0.972
12_Marcus	5/5	5/5	5/5	5/5	5/5	5/5	5/5	1.000
13_Nat	6/6	5/5	5/5	5/5	4/5	5/6	5/5	0.946
14_StephenV2	5/5	4/5	5/5	5/5	5/5	5/5	5/5	0.971
16_Bella	5/5	3/5	5/5	5/5	5/5	5/5	5/5	0.943
17_Raquel	5/5	4/5	4/5	5/5	4/5	5/5	5/5	0.914
1_Alan	5/5	5/5	5/5	5/5	5/5	5/5	5/5	1.000
2_Alex	5/5	5/5	5/5	5/5	1/5	5/5	5/5	0.886
3_Anghad	4/5	1/5	5/5	5/5	5/5	5/5	5/5	0.857
4_Daniel	5/5	3/5	5/5	5/5	5/5	5/5	5/5	0.943
5_Georgia	5/5	5/5	5/5	4/5	5/5	5/5	5/5	0.971
6_Harry	5/5	3/5	5/5	5/5	4/5	5/5	5/5	0.914
7_Henry	5/5	4/5	5/5	5/5	5/5	1/5	5/5	0.857
8_Jestin	6/6	5/5	5/5	5/5	5/5	5/5	5/5	1.000
Stephen	2/20	11/21	17/20	16/20	17/20	16/20	18/20	0.688
Val	2/5	3/5	5/5	5/5	4/5	5/5	5/5	0.829
MEAN	0.843	0.796	0.979	0.965	0.897	0.931	0.994	0.915